

Pencils to Pixels: Reimplementing SDEdit

by: Rio Voss-Kernan (rvossker), William Jarvis (wjarvis10), and John Perdue (jperdue21)
CSCI 1470 (Deep Learning), Department of Computer Science, Brown University

Introduction

The ability to manipulate digital images with precision and realism has become increasingly important in fields ranging from design to deep learning. One of the key challenges is finding techniques that can modify an image while preserving its underlying structure and coherence. Traditional generative models including GANs and unconditional diffusion models, struggle to balance faithfulness to the input and realism. To address this, we explore a method called SDEdit, which enables controlled image editing by hijacking the denoising property of diffusion models with a artificially noised input input. Our implementation is based on the paper Image Editing with SDEdit: Guided Image Synthesis via Stochastic Differential Equations, and leverages a pretrained diffusion model for image generation of flowers to perform meaningful and accurate edits.

Methodology

We implemented SDEdit by leveraging a pretrained diffusion model trained on the Oxford Flowers 102 dataset. Starting from an input sketch, we added Gaussian noise corresponding to a selected diffusion time step to simulate a partially noised image. We then used the reverse diffusion process to denoise this image, guiding the generation toward photorealistic outputs while preserving faithfulness to the original input. The pretrained model's denoising function was applied iteratively, gradually refining the noisy input into a realistic image. Our implementation closely follows the SDEdit framework, ensuring controlled and faithful edits without retraining the diffusion model.

Results

The plots on the left display two examples of sketches across different t_0 values and the realism/faithfulness associated with them. In both instances, we saw the L2 Distance (Our Measure for Faithfulness) increase with higher t_0 , with values ranging from 0-0.25 in the sweet spot. This is intuitive as with higher t_0 , the output is likely to be less faithful to the original sketch. Conversely, we saw KID (Our Measure for Realism) fall as t_0 increased. The sweet spot values differed by sketch, but ranged from 15-30.

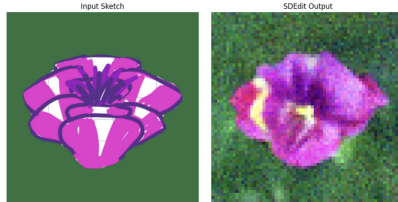


Figure 1. Full generation from sketch noised to $t=0.35$

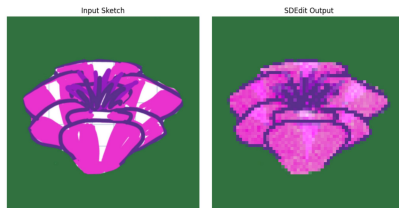


Figure 2. Masked "edit" from sketch noised to $t=0.4$. Mask included background and petal outlines

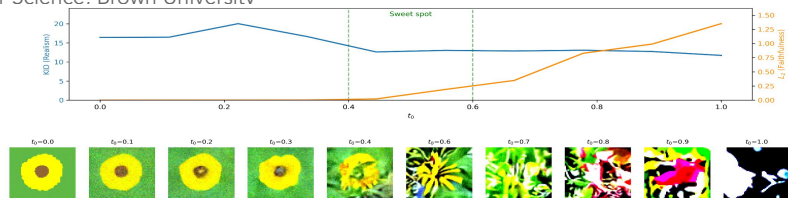


Figure 3: Faithfulness/Realism curve over varying levels of input noise. "Sweet Spot" found to be $t = 0.4-0.6$. (Faithfulness measured by L2 difference and Realism by Kernel Inception Distance)

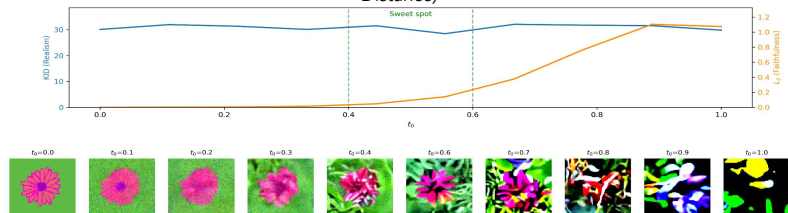


Figure 4. Faithfulness/Realism curve over varying levels of input noise. "Sweet Spot" found to be $t = 0.4-0.6$. (Faithfulness measured by L2 difference and Realism by Kernel Inception Distance)

Discussion

While our results did not match the original paper, they did show the Realism/Faithfulness tradeoff that is core to this problem. As increasing t_0 allows more significant denoising at the beginning of our process, it results in the final output having deviated more from the original sketch (Shown by the increase in L2 distance). Conversely, if we do not denoise the image enough, we sacrifice realism in the output (Shown by the higher KID at lower t_0 values).

One issue we encountered in this process was evaluating the KID. L2 Distance was much simpler, as we were comparing the sketch to the output. When measuring KID, we are comparing the output to the real image we attempted to sketch, so the results are somewhat dependent on the quality of the sketch. Additionally, KID does not allow an image to image comparison, and requires multiple outputs to compare to the one real image. We only utilized 3 samples because of the time it takes to create 10 edited images, but lacking sample size could be another reason our realism curve is not smoother and clearly decreasing, as shown in the original paper.